

COMPSCI 689

Lecture 2: Linear Regression

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A definition of machine learning



Mitchell (1997): “A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P , if its performance at tasks in T , as measured by P , improves with experience E .”

Substitute “training data D ” for “experience E .”

Supervised Learning

- In the supervised setting, the data take the form of tuples $(\mathbf{x}, \mathbf{y})$.
- In this setting, the $\mathbf{x}$'s are readily observable.
- However, for some reason, the $\mathbf{y}$'s are not readily observable, but can be obtained at some cost.
- Thus, given the value of $\mathbf{x}$, we would like to *predict* the corresponding value of $\mathbf{y}$.
- To predict the value of $\mathbf{y}$ that corresponds to a given $\mathbf{x}$, we use a *prediction function* $f(\mathbf{x})$.
- The two main types of supervised problems are regression where $\mathbf{y}$ is real-valued, and classification where $\mathbf{y}$ is categorical.
- Examples?

General Supervised Learning Notation

- Input Space: $\mathcal{X}$
- Output Space: $\mathcal{Y}$
- Input: $\mathbf{x} \in \mathcal{X}$
- Output: $\mathbf{y} \in \mathcal{Y}$
- Prediction Function: $f: \mathcal{X} \rightarrow \mathcal{Y}$

Example: Binary Classification

In binary classification, the output y can be one of two possible values. The output y is referred to as the “class” or “label” of $\mathbf{x}$. The inputs $\mathbf{x}$ are also referred to as “features.” The goal is to predict the class given the features.

- Input Space: $\mathcal{X}$
- Output Space: $\mathcal{Y} = \{0, 1\}$ or $\{-1, 1\}$
- Input: $\mathbf{x} \in \mathcal{X}$
- Output: $y \in \mathcal{Y}$
- Prediction Function: $f: \mathcal{X} \rightarrow \mathcal{Y}$

Example: Multi-Class Classification

In multi-class classification, the output y can be one of C possible values. The output y is referred to as the “class” or “label” of $\mathbf{x}$. The inputs $\mathbf{x}$ are also referred to as “features.” The goal is to predict the class given the features.

- Input Space: $\mathcal{X}$
- Output Space: $\mathcal{Y} = \{1, \dots, C\}$
- Input: $\mathbf{x} \in \mathcal{X}$
- Output: $y \in \mathcal{Y}$
- Prediction Function: $f: \mathcal{X} \rightarrow \mathcal{Y}$

Example: Regression

In regression, the output y is a real-valued number also called the “target,” “dependent variable,” “outcome variable” or “response variable.” The $\mathbf{x}$ inputs are also called “features,” “independent variables,” “predictors” or “covariates.”

- Input Space: $\mathcal{X}$
- Output Space: $\mathcal{Y} = \mathbb{R}$
- Input: $\mathbf{x} \in \mathcal{X}$
- Output: $y \in \mathcal{Y}$
- Prediction Function: $f: \mathcal{X} \rightarrow \mathcal{Y}$

The Supervised Learning Problem

The Supervised Learning Problem

Given a *data set* consisting of a collection of input-output tuples $\mathcal{D} = \{(\mathbf{x}_n, \mathbf{y}_n) | \mathbf{x}_n \in \mathcal{X}, \mathbf{y}_n \in \mathcal{Y}, 1 \leq n \leq N\}$, select the best prediction function $f: \mathcal{X} \rightarrow \mathcal{Y}$.

Note: A data set is not a mathematical set. It is a collection of elements that allows repetition.

Prediction Loss Functions

Prediction Loss Function: A prediction loss function

$L: \mathcal{Y} \times \mathcal{Y} \rightarrow \mathbb{R}$ is a real-valued function that is bounded below (typically at 0), and that satisfies $L(\mathbf{y}, \mathbf{y}) \leq L(\mathbf{y}, \mathbf{y}')$ for all $\mathbf{y}, \mathbf{y}' \in \mathcal{Y}$.

Examples:

- Squared Loss: $L_{sq}(\mathbf{y}, \mathbf{y}') = \|\mathbf{y} - \mathbf{y}'\|_2^2 = \sum_{k=1}^K (\mathbf{y}_k - \mathbf{y}'_k)^2$
- Absolute Loss: $L_{abs}(\mathbf{y}, \mathbf{y}') = \|\mathbf{y} - \mathbf{y}'\|_1 = \sum_{k=1}^K |\mathbf{y}_k - \mathbf{y}'_k|$
- 0/1 Loss: $L_{01}(\mathbf{y}, \mathbf{y}') = [\mathbf{y} \neq \mathbf{y}']$

Given a prediction loss function L , an instance $(\mathbf{x}, \mathbf{y})$, and a prediction function f , we compute the loss of f on $(\mathbf{x}, \mathbf{y})$ as $L(\mathbf{y}, f(\mathbf{x}))$.

Do we now have enough information to select the optimal f given a data set $\mathcal{D}$?

Prediction Function Models

- In general in supervised learning, we do not attempt to identify the best function f from the set of all possible functions mapping from $\mathcal{X}$ to $\mathcal{Y}$.
- Instead, we specify a specific set of functions $\mathcal{F}$ and select from that set.
- We will refer to the set $\mathcal{F}$ as a *prediction function model* or just a *model*.
- The set $\mathcal{F}$ can be finite, but it is more typically uncountably infinite.

Do we now have enough information to select the optimal f given a data set $\mathcal{D}$?

Empirical Risk Minimization

Let $\mathcal{F}$ be a set of functions mapping from $\mathcal{X}$ to $\mathcal{Y}$ (e.g., a prediction function model). The principle of Empirical Risk Minimization (ERM) states that we should select the function f from the set $\mathcal{F}$ that *minimizes the average of the prediction loss* $L(\mathbf{y}_n, f(\mathbf{x}_n))$ computed over the data set $\mathcal{D}$, also known as the empirical risk $R(f, \mathcal{D})$:

$$\hat{f} = \arg \min_{f \in \mathcal{F}} R(f, \mathcal{D})$$

$$R(f, \mathcal{D}) = \frac{1}{N} \sum_{n=1}^N L(\mathbf{y}_n, f(\mathbf{x}_n))$$

Supervised Learning by ERM

ERM provides our first general framework for supervised learning:

- 1 The supervised learning task defines $\mathcal{X}$ and $\mathcal{Y}$.
- 2 We collect or obtain a data set $\mathcal{D}$.
- 3 We choose a prediction loss function L as the performance measure.
- 4 We choose the space of prediction functions $\mathcal{F}$.
- 5 We select the function $\hat{f}$ from $\mathcal{F}$ that minimizes the empirical risk $R(f, \mathcal{D})$.

Linear Regression

- Consider the classical regression setting in which $\mathcal{X} = \mathbb{R}^D$ and $\mathcal{Y} = \mathbb{R}$.
- In this setting, the data set $\mathcal{D}$ consists of input vectors $\mathbf{x}_n$ and scalar output values y_n .
- We will assume that $\mathbf{x}_n$ is a row vector, and thus has shape $(1, D)$.
- In linear regression, we choose as our model $\mathcal{F}$ the space of all linear functions of $\mathbf{x}$.
- The most commonly used prediction loss function in this setting is the squared loss $L_{sq}(y, y') = (y - y')^2$.

The Linear Regression Model

- To apply ERM to the linear regression model, we need a mathematical description of the set $\mathcal{F}$ of all linear functions of $\mathbf{x}$.
- First, define the parameter space $\Theta = \{[\mathbf{w}; b] | \mathbf{w} \in \mathbb{R}^D, b \in \mathbb{R}\}$.
- An element $\theta \in \Theta$ is a vector $\theta = [\mathbf{w}; b]$, referred to as the *model parameters*.
- $\mathbf{w}$ is a column vector with shape $(D, 1)$ called the *weights* or *coefficients* and b is a real scalar called the *bias* or *intercept*.
- Now define the set of parametric functions $\mathcal{F} = \{f_{\theta}(\mathbf{x}) = \mathbf{x}\mathbf{w} + b | \theta \in \Theta\}$.
- This parametric space of functions is the linear regression prediction function model.

ERM for Linear Regression

Given the choice of the squared prediction loss $L_{sqr}(y, y') = (y - y')^2$ and the space of prediction functions $\mathcal{F} = \{f_{\theta}(\mathbf{x}) = \mathbf{x}\mathbf{w} + b | \theta \in \Theta\}$, we can now apply ERM to define the optimal prediction function:

$$\hat{f} = \arg \min_{f \in \mathcal{F}} R(f, \mathcal{D}) \rightarrow \hat{\theta} = \arg \min_{\theta \in \Theta} R(f_{\theta}, \mathcal{D})$$

$$R(f, \mathcal{D}) = \frac{1}{N} \sum_{n=1}^N (\mathbf{y}_n - f(\mathbf{x}_n))^2 \rightarrow R(f_{\theta}, \mathcal{D}) = \frac{1}{N} \sum_{n=1}^N (\mathbf{y}_n - (\mathbf{x}_n \mathbf{w} + b))^2$$

ERM for Linear Regression

Given the choice of the squared prediction loss $L_{sq}(y, y') = (y - y')^2$ and the space of prediction functions $\mathcal{F} = \{f_\theta(\mathbf{x}) = \mathbf{x}\mathbf{w} + b | \theta \in \Theta\}$, we can now apply ERM to define the optimal prediction function:

$$\hat{\theta} = \arg \min_{\theta \in \Theta} R(f_\theta, \mathcal{D})$$

$$R(f_\theta, \mathcal{D}) = \frac{1}{N} \sum_{n=1}^N (\mathbf{y}_n - (\mathbf{x}_n \mathbf{w} + b))^2$$

Question: How do we actually find the model parameters θ that minimize the empirical risk defined above?

Optimization Theory for ERM

Key optimization definitions for minimizing empirical risk:

- **Gradient:** The gradient $\nabla R(f_\theta, \mathcal{D})$ of the empirical risk is the vector of partial derivatives of $R(f_\theta, \mathcal{D})$ with respect to each of the model parameters: $[\nabla R(f_\theta, \mathcal{D})]_i = \frac{\partial}{\partial \theta_i} R(f_\theta, \mathcal{D})$.
- **Hessian:** The hessian $\nabla^2 R(f_\theta, \mathcal{D})$ of the empirical risk $R(f_\theta, \mathcal{D})$ is the matrix of mixed partial derivatives of $R(f_\theta, \mathcal{D})$ with respect to each pair of model parameters:
$$[\nabla^2 R(f_\theta, \mathcal{D})]_{ij} = \frac{\partial^2}{\partial \theta_i \partial \theta_j} R(f_\theta, \mathcal{D}).$$
- **Local Minimizer:** θ is a local minimizer of $R(f_\theta, \mathcal{D})$ if and only if $\nabla R(f_\theta, \mathcal{D}) = 0$ and the Hessian of $R(f_\theta, \mathcal{D})$ at θ is positive semi-definite (PSD).
- **Global Minimizer:** θ is a global minimizer of $R(f_\theta, \mathcal{D})$ if θ is a local minimizer of $R(f_\theta, \mathcal{D})$ and $R(f_\theta, \mathcal{D}) \leq R(f_{\theta'}, \mathcal{D})$ for all $\theta' \in \Theta$.

Closed-Form Optimization Recipe for ERM

- 1 Derive the gradient $\nabla R(f_\theta, \mathcal{D})$.
- 2 Solve the gradient equation $\nabla R(f_\theta, \mathcal{D}) = 0$, obtaining all solutions.
- 3 Determine which solutions of the gradient equation are local minimizers by checking the hessian condition.
- 4 Check the value $R(f_\theta, \mathcal{D})$ at each local minimizer to determine which are global minimizers.

Helpful Results

Some helpful results for optimizing the linear regression model.

- **Bias Absorption:** The prediction function $\mathbf{x}\mathbf{w} + b$ can be expressed as a single inner product by defining $\tilde{\mathbf{x}} = [\mathbf{x}, 1]$. We then have $\tilde{\mathbf{x}}\theta = \mathbf{x}\mathbf{w} + b$. For simplicity, we will assume bias absorption and write the prediction function as $\mathbf{x}\theta$.
- **Matrix form of the Risk:** The empirical risk function is easier to work with in matrix form. Define $\mathbf{X}$ to be the $N \times (D + 1)$ matrix of (bias absorbed) inputs and $\mathbf{Y}$ to be the $N \times 1$ matrix of outputs. Then:

$$\frac{1}{N} \sum_{n=1}^N (\mathbf{y}_n - \mathbf{x}_n\theta)^2 = \frac{1}{N} (\mathbf{Y} - \mathbf{X}\theta)^T (\mathbf{Y} - \mathbf{X}\theta)$$

- **Matrix Calculus:** We will need two basic matrix calculus results: $\nabla \mathbf{c}^T \theta = \mathbf{c}$ where $\mathbf{c}$ is a $(D + 1) \times 1$ vector, and $\nabla \theta^T \mathbf{A} \theta = 2\mathbf{A}\theta$ for $\mathbf{A}$ a $(D + 1) \times (D + 1)$ real symmetric matrix.

Step 1: Derive Gradient

$$\begin{aligned}\nabla R(f_{\theta}, \mathcal{D}) &= \nabla \frac{1}{N} (\mathbf{Y} - \mathbf{X}\theta)^T (\mathbf{Y} - \mathbf{X}\theta) \\&= \frac{1}{N} \nabla (\mathbf{Y}^T - \theta^T \mathbf{X}^T) (\mathbf{Y} - \mathbf{X}\theta) \\&= \frac{1}{N} \nabla (\mathbf{Y}^T \mathbf{Y} - \mathbf{Y}^T \mathbf{X}\theta - \theta^T \mathbf{X}^T \mathbf{Y} + \theta^T \mathbf{X}^T \mathbf{X}\theta) \\&= \frac{1}{N} \nabla (\mathbf{Y}^T \mathbf{Y} - 2\mathbf{Y}^T \mathbf{X}\theta + \theta^T \mathbf{X}^T \mathbf{X}\theta) \\&= \frac{1}{N} (-2\mathbf{X}^T \mathbf{Y} + 2\mathbf{X}^T \mathbf{X}\theta)\end{aligned}$$

Step 2: Solve Gradient Equation

$$\begin{aligned}\nabla R(f_\theta, \mathcal{D}) &= 0 \\ \Rightarrow \frac{1}{N}(-2\mathbf{X}^T \mathbf{Y} + 2\mathbf{X}^T \mathbf{X} \theta) &= 0 \\ \Rightarrow \frac{2}{N} \mathbf{X}^T \mathbf{X} \theta &= \frac{2}{N} \mathbf{X}^T \mathbf{Y} \\ \Rightarrow \mathbf{X}^T \mathbf{X} \theta &= \mathbf{X}^T \mathbf{Y} \\ \Rightarrow \hat{\theta} &= (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y}\end{aligned}$$

Note: This solution is only well-defined if $\mathbf{X}^T \mathbf{X}$ is invertible! It will be invertible if it is strictly positive definite.

Step 3: Check Hessian

- It's easy to see that $\nabla^2 R(f_\theta, \mathcal{D}) = \frac{2}{N} \mathbf{X}^T \mathbf{X}$, which is PSD.
- Thus, the solution $\hat{\theta}$ is a local minimizer if it is well-defined.
- Since there is at most one local minimizer $\hat{\theta}$, $\hat{\theta}$ is the global minimizer so long as it is well defined.
- Therefore, $\hat{\theta}$ is the solution to the ERM learning problem for the linear regression model with squared loss when it is well defined.

Making Predictions

- To make a prediction for a new data point $\mathbf{x}_*$, we compute:

$$\hat{y} = f_{\hat{\theta}}(\mathbf{x}_*) = \mathbf{x}_* \hat{\theta}$$